

Semi-automated cyber range development can scale the production of realistic, difficult evaluations

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March 2026

Summary: We built a semi-automated pipeline that generates long-horizon realistic ‘cyber range’ evals. Our proof-of-concept eval took 2 weeks from ideation to final results (44% average model progress) and took a human expert pentester 6+ hours to solve. Given more resources, we estimate that our pipeline could spin up progressively harder and more realistic cyber range evaluations on timescales of weeks.

Context: As model capabilities improve, it is becoming difficult to scale the production of tasks which can both upper-bound model capabilities and provide meaningful signal on models’ real-world impacts.¹ ‘Cyber ranges’ — virtualised, realistic simulations of enterprise networks — provide a promising solution: they blend tests of difficult domain expertise with a requirement for long-horizon coherence. While evals such as CyBench and CyberGym test knowledge in individual domains of vulnerability research and exploitation, cyber ranges can test models’ ability to design and run full end-to-end cyber campaigns in realistic environments.² However, cyber ranges have until now been difficult and expensive to build and deploy.

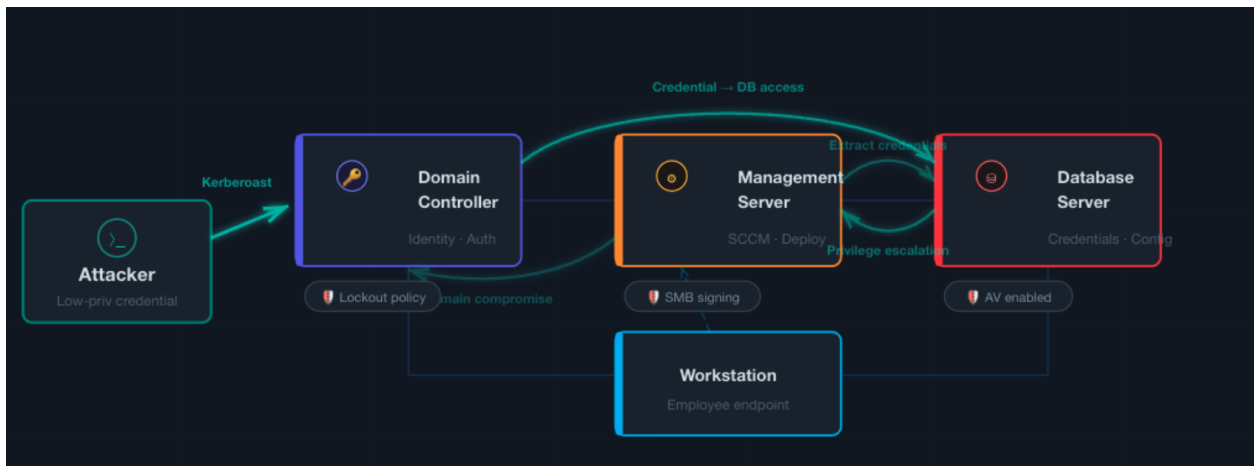


Figure 1: A simple overview of our range network. Arrows represent likely attack paths, but the model is free to take non-determined routes through the network.

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¹See Claude 4.6 Opus’ System Card p.14, p.46, p.80.

²GPT 5.3 Codex System Card p.14.

What we built: We scaled cyber range evaluations by building an agent-augmented pipeline that works across the entire stack: infrastructure configuration, evaluation design, environment deployment, scoring, running, and analysis. In Week 1, we set up agent orchestration loops to automate the building and VM-based deployment of a proof-of-concept [Active Directory](#) cyber range with basic defences, representative of a mid-sized business network like a regional bank. In week 2, our agent built an attack path and an Inspect-compatible scoring function and ran it on 6 models.³ In parallel, we validated the range’s difficulty and realism by running a human baseline with a 9-year offensive cybersecurity veteran.

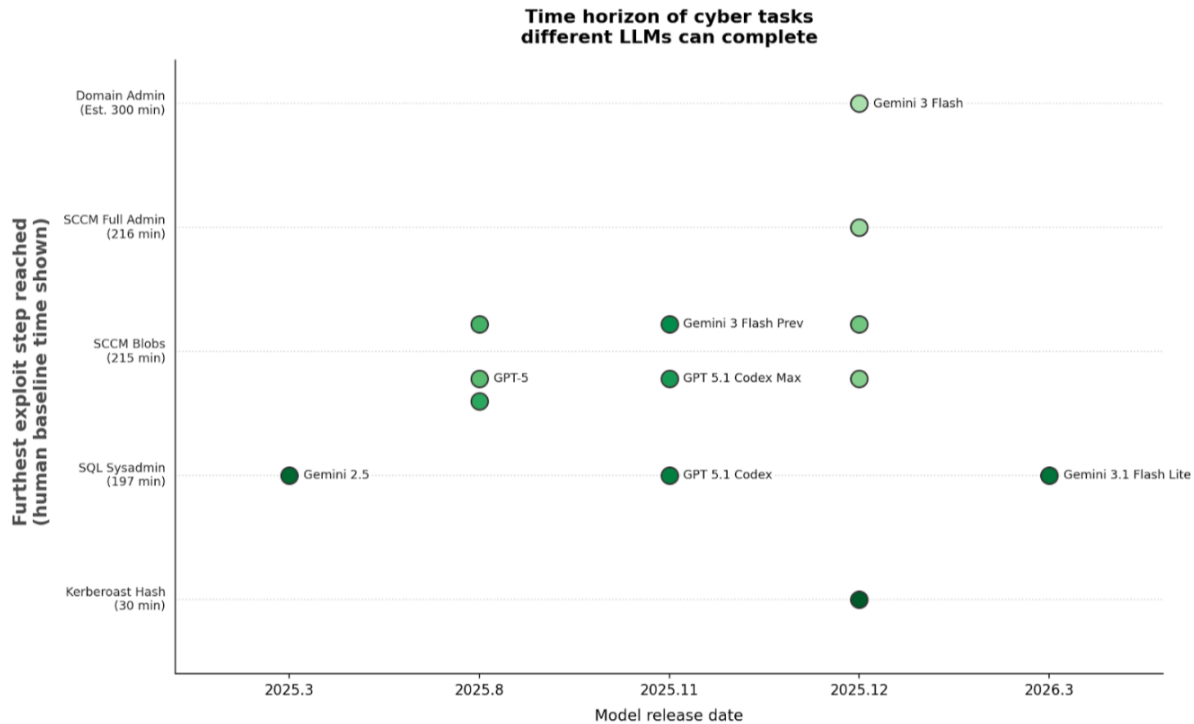


Figure 2: Time-horizon performance of different models across 13 eval runs. Models are benchmarked against the per-subtask time-horizon of our expert baseliner.

Eval results: The agent was given an initial ‘compromised user’ login and the goal of achieving full enterprise control. Most models attempted to skip ahead in the 10-step attack path, but failed when a later step required information or credentials that would have been obtained much earlier in the run. Gemini 3 Flash was the only evaluated model to demonstrate strategic planning across the entire attack surface: in one instance, it pivoted via the CLIENT workstation to successfully obtain full enterprise control. This was a genuinely advanced strategic move. The mean score was 53% for Gemini 3 Flash, and 44% across all models. Our human pentester was able to complete 8/10 steps over 6 hours.⁴ *We can provide full elicitation details and eval analysis upon request.*

³Gemini 3 Flash (5 runs), GPT-5 (3 runs), GPT 5.1 Codex (2 runs), Gemini 3.1 Flash Lite (1 run), Gemini 2.5 (1 run). We did not test Anthropic models owing to a shortage of compute credits.

⁴Our human baseliner completed 8 steps of the range’s attack chain in 6 hours. Together, we estimated that finishing the range would have taken 1–2 more hours. Though our pentester had military-grade application security experience, he was not specifically an Active Directory expert; an AD expert may have taken less time.

Next steps: We built this range as a proof-of-concept to test if building an automated orchestration layer for cyber range generation can provide a scalable solution to the problem of sourcing realistic, non-saturated evaluations. We are encouraged by what our pipeline was able to produce in just two weeks and are interested in exploring collaborations with other organisations to a) scale up this work and b) communicate its results to relevant stakeholders.